



Technology Data Sheet (TDS)

TINMORRY ABS-CF

Version 1.0

• Basic Introduction

TINMORRY ABS-CF is a carbon fiber reinforced ABS composite. Under the reinforcement of carbon fiber, it has significantly higher strength and stiffness than normal ABS, and can greatly improve the warping problem of the prints, thus endow the products with higher dimensional stability and accuracy. In addition, it has higher heat and wear resistance than normal ABS, allowing the products to meet higher using requirements.

Note:1. Since the ABS-CF filament contains carbon fiber, please use a nozzle with a diameter of ≥ 0.4 mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them. 2. It is more recommended to use an enclosed printer for printing, especially for large-sized and high-density models, so as to reduce warping and ensure sufficient layer strength. 3. It will produce a pungent odor during printing, please ensure good ventilation.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g $\pm$ 0.5%
Length	380 ± 5 m
Spool material	Cardboard (Temperature resistance: 150 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	70 - 80 °C, 6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	$\leq 65\text{ °C}$, $\leq 20\%$ RH (Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set the temp to be about 60 °C)
Storage temperature and humidity	$\leq 30\text{ °C}$, $\leq 20\%$ RH (Sealed with desiccant)
Compatible support material	Itself or support filament for ABS
Compatible printer type	Enclosed-frame (recommended) Open-frame (not recommended)
Compatible nozzle material	Hardened steel, tungsten carbide, etc
Compatible nozzle size	0.4, 0.6 (recommended), 0.8 mm
Compatible plate type	Smooth / Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	260 - 280 °C
Bed temperature	90 - 100 °C (depending on the printer and plate types)
Printing speed*	Max: 300 mm/s
Chamber temperature	$< 65\text{ °C}$

*When printed with Bambu X1C and P1S, and the nozzle temperature is 275 °C, the max printing speed can reach 300 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY ABS-CF have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.09 g/cm ³
Saturated water absorption rate	25 °C , 55% RH	0.47%
Melt index	260 °C, 2.16 kg	7.3 ± 1.4 g/10 min
Glass transition temperature	DSC, 10 °C/min	N / A
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	206 °C
Vicar softening temperature	ISO 306, GB/T 1633	111 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	91 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	100 °C

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	37 ± 3 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	28 ± 3 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	2823 ± 159 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2158 ± 149 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	$3.2\% \pm 1.2\%$
Breaking elongation rate (Z)	ISO 527, GB/T 1040	$1.1\% \pm 0.7\%$
Bending strength (XY)	ISO 178, GB/T 9341	72 ± 2 MPa
Bending strength (Z)	ISO 178, GB/T 9341	55 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	2876 ± 126 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	2241 ± 116 MPa
Impact strength (XY)	ISO 179, GB/T 1043	15.6 ± 0.8 kJ/m ²
Impact strength (Z)	ISO 179, GB/T 1043	8.7 ± 1.1 kJ/m ²

• Chemical and Other Properties

Subjects	Data
Component	ABS, carbon fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Not resistant to some kinds oil and grease for long-term contact
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Good
Resistance to strong acid	Not resistant
Resistance to weak alkali	Good
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Pungent
Odor during combustion	Strongly Pungent

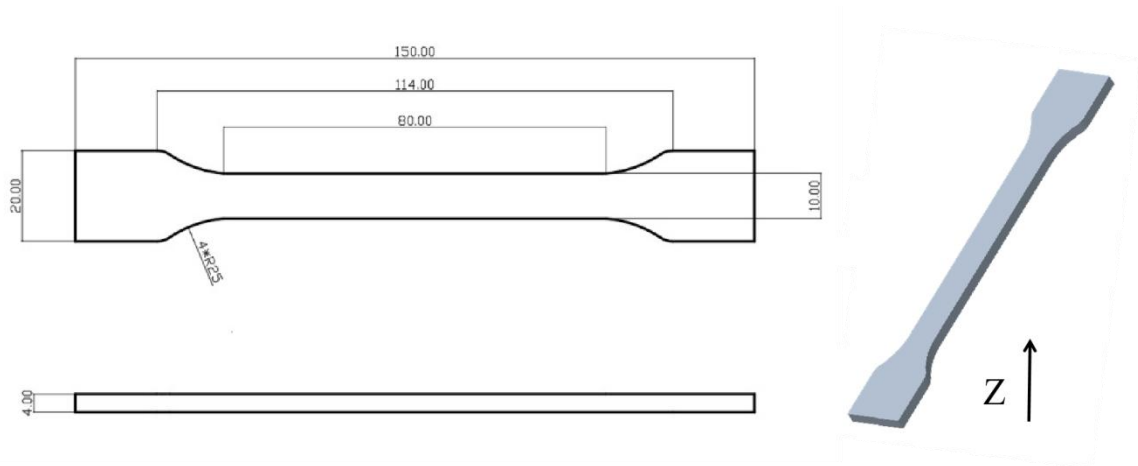
• Specimen Info

Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	275 °C
Bed temperature	100 °C
Printing speed	XY: 150 mm/s; Z: 100 mm/s
Infill density	100%
Infill pattern	Concentric

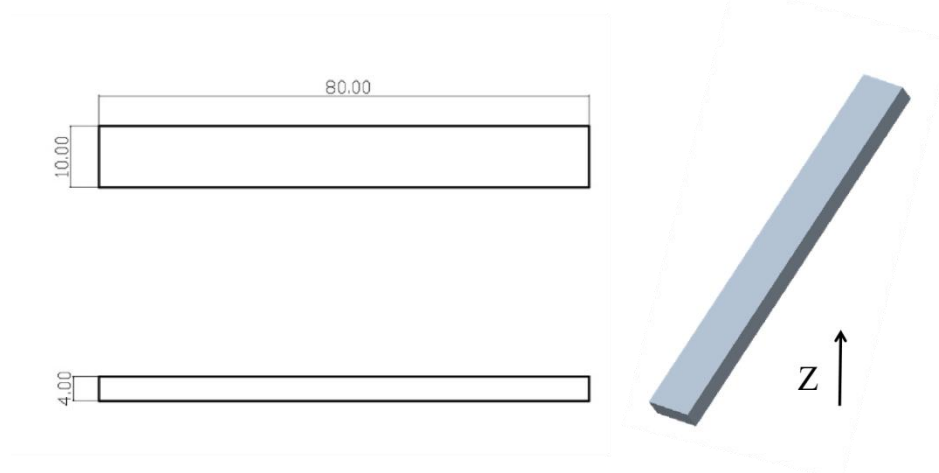
Statements:

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties and heat resistance of TINMORRY ABS-CF prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.

1. Tensile Test



2. Bending Test and Impact Test



- **Disclaimer**

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them, or the usage scenarios of them.